

Topic 1: Biological Molecules

This topic introduces the chemicals of life: the organic and inorganic molecules and ions that are fundamental to the structure and physiology of living organisms. The role of monomers in the synthesis of polymers and how the structure and properties of these relate to their functions are considered. An understanding of scientific method is developed in the practical investigation of enzyme action.

Opportunities for developing mathematical skills within this topic include: recognising and making use of appropriate units in calculations; using ratios, fractions and percentages; constructing and interpreting frequency tables and diagrams; translating information between graphical, numerical and algebraic forms; understanding that $y = mx + c$ represents a linear relationship; determining the intercept of a graph; calculating rate of change from a graph; drawing and using the slope of a tangent to a curve as a measure of rate of change. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Students should:

1.1 Carbohydrates

- i Know the difference between monosaccharides, disaccharides and polysaccharides.
- ii Know the structure of the hexose glucose (alpha and beta) and the pentose ribose.
- iii Understand how monosaccharides (glucose, fructose, galactose) join to form disaccharides (sucrose, lactose and maltose) and polysaccharides (starch formed from amylose and amylopectin; glycogen) through condensation reactions forming glycosidic bonds, and how these can be split through hydrolysis reactions.
- iv Understand how the structure of glucose, starch, glycogen and cellulose relates to their function.